EXERCIZE 4.

1. SELECT order_id, customer_id, order_date, DAYNAME (order_date) AS day_name
FROM orders;

Table orders

order_id	customer_id	order_date	day_name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

2. SELECT customer_id, customer_name, sign-up_date, MONTHNAME (signup_date) AS signup_month_name
FROM customer_signups;

Table customer_signups

customer_id	customer_name	sign-up_date	signup_month_name
201	John	2026-01-15	January
202	Mary	2026-02-20	february
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May.

3. SELECT sale_id, product_name, sale_date, MONTH(sale_date) AS sale_month
FROM sales;

Table sales;

sale_id	product_name	sale_date	sale_month
S001	laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

4. SELECT transaction_id, customer_id, transaction_date, YEAR(transaction_date) AS transaction_year
FROM transactions;

Table transactions.

transaction_id	customer_id	transaction_date	transaction_year.
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

5. SELECT delivery_id, customer_id, delivery_date, DAY(delivery_date) AS day_of_month.
FROM deliveries

Table deliveries

delivery_id	customer_id	delivery_date	day_of_month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

6. SELECT employee_id, employee_name, department, CURRENT_DATE() AS today_date
FROM employees;

Table employees

employee_id	employee_name	department	today_date
301	Nandi	Sales	2026-06-18
302	Brian	IT	2026-06-18
303	Lerato	Finance	2026-06-18
304	Sipho	HR	2026-06-18
305	Aisha	Marketing	2026-06-18

7. SELECT order_id, customer_id, order_date_text, TO_DATE(order_date_text) AS order_date
From online_orders;

Table online_orders;

order_id,	customer_id	order_date_text	order_date
4001	101	2026-June 15	2026-01-15
4002	102	2026 Feb 20	2026-02-20
4003	103	2026 Mar 25	2026-03-25
4004	104	2026 April 10	2026-04-10
4005.	105	2026 May 25	2026-05-05

8. SELECT payment_id, customer_id, payment_date,
TO_CHAR(payment_date; 'YYYY-MM-DD') AS formatted_payment_date
FROM payment_dates;

Table payment_dates

Payment_id	customer_id	payment_date	formatted_payment_date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

9. SELECT customer_id, customer_name, last_purchase_date,
DATEDIFF ~~[illegible]~~ (CURRENT_DATE( ), last_purchase_dat) AS days_since_last_purchase

FROM customer_purchases

Table customer_purchases;

customer_id	customer_name	last_purchase_date	days_since_last_purchase
501	John	2026-05-01	48
502	Mary	2026-05-10	49
503	Peter	2026-05-15	34
504	Sarah	2026-05-20	29
505	Thabo	2026-05-25	24

10. SELECT order_id, customer_id, order_date,
DATEADD(order_date, INTERVAL 7 DAY) AS expeted_delivery_date
FROM shipping_orders

Table shipping_orders

order_id	customer_id,	order_date	expeted_delivery_date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

11. SELECT booking_id, customer_id, booking_date,
YEAR(Booking_date) AS booking_year,
MONTH (Booking_date) AS booking_month,
DAY(booking_date) AS booking_day
FROM bookings;

Table bookings

booking_id	customer_id	booking_date	booking_year	booking_month	Booking_day
B001	101	2026-01-12	2026	~~12~~ 1	12
B002	102	2026-02-18	2026	~~18~~ 2	18
B003	103	2026-03-22	2026	~~22~~ 3	22
B004	104	2026-04-09	2026	~~9~~ 4	9
B005	105	2026-05-27	2026	~~27~~ 5	27

12. SELECT order_id, customer_id, order_date,
YEAR(order_date) AS order_year, amount
FROM YEARLY_ORDERS
WHERE YEAR(order_date) = 2026;

Table yearly_orders.

Order_id	customer_id	order_date	order_year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

13. SELECT order_id, customer_id, order_date,
MONTH(order_date) AS order_month, amount
FROM monthly_orders
WHERE MONTH(order_date) = 3;

Table monthly_orders

order_id	customer_id	order_date	order_month	amount
8003	103	2026-03-10	3	800
8004	104	2026-03-25	3	1500

14. SELECT subscription_id, customer_id, start_date,
LAST_DAY(start_date) AS month_end_date
FROM subscriptions

Table subscriptions

subscription_id	customer_id	start_date	month_end_date
SUB 001	101	2026-01-10	2026-01-31
SUB 002	102	2026-02-15	2026-02-28
SUB 003	103	2026-03-20	2026-03-31
SUB 004	104	2026-04-25	2026-04-30
SUB 005	105	2026-05-30	2026-05-31

15. SELECT send_id, customer_id, send_date,
DATE_TRUNC(send_date, MONTH) As month_start_date
FROM campain_sends;

Table campain_sends

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-31	2026-04-01
C005	105	2026-05-27	2026-05-01

16. SELECT invoice_id, customer_id, invoice_date,
TO_CHAR(invoice_date, 'Month YYYY') AS invoice_month_year
FROM invoice_dates;

Table invoice_dates.

invoice_id	customer_id	invoice_date	invoice_month_year.
INV 001	101	2026-01-05	January 2026
INV 002	102	2026-02-10	February 2026
INV 003	103	2026-03-15	March 2026
INV 004	104	2026-04-20	April 2026
INV005	105	2026-05-20	May 2026

17. SELECT customer-id, customer_name, date_of_birth,
        TIMESTAMPDIFF(YEAR, date_of_birth, CURRENT_DATE()) AS customer_age
    FROM customer_birthdays;

Table customer_birthdays

customer_id	customer_name	date_of_birth	customer_age
901	John	1998-05-10	28
902	Mary	1990-08-20	35
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
905	Thabo	2000-07-30	25

18. SELECT order_id, customer_id, order_date,
        DAYNAME(order_date) AS day_name,
        CASE
            WHEN DAYNAME(order_date) IN ('Sartuday', 'Sunday') THEN 'weekend'
        ELSE 'WEEKday'
    END AS day_type
FROM weekend_orders;

Table weekend_orders

order_id	customer_id	order_date	day_name	day_type
9001	101	2026-05-01	Friday	Weekday
9002	102	2026-05-02	Sartuday	Weekend
9003	103	2026-05-03	Sunday	Weekend
9004	104	2026-05-04	Monday	Weekday
9005	105	2026-05-05	Tuesday	Weekday

19. SELECT transaction_id, customer_id, transaction_date,
        QUATER(transaction_date) AS transaction_quarter, amount
FROM quartely_transactions

Table quarterly_transactions

transaction_id	customer_id	transaction_date	transaction_quarter	amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-20	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

20.
```
SELECT order_id, customer_id, order_date,
       DATEDIFF(CURRENT_DATE( ), order_date) AS days_since_order, amount
FROM recent_orders
WHERE DATEDIFF(CURRENT( ), order_dat) >30;
```

Table recent_orders

order_id	customer_id	order_date	date_since_order	amount
R001	101	2026-04-01	78	500
R002	102	2026-04-15	64	1200
R003	103	2026-05-01	48	800

BONUS:
```
SELECT customer_id, customer_name, last_purchase_date,
    DATEDIFF (CURRENT_DATE (), last_purchase_date) AS days_since_last_purchase,
    CASE
      WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date <=30 THEN
      'Active customer'
      WHEN DATEDIFF(CURRENT_DATE( ), last_purchase_dateBETWEEN 31 AND 90
      THEN 'At Risk Customer'
    ELSE 'Inactive Customer'
  END AS customer_status
FROM customer_recency;
```

Table recency

customer_id	customer_name	last_purchase_date	days_since_last_purchase	customer_status
1001	John	2026-05-25	24	Active customer
1002	Mary	2026-05-10	39	At risk customer
1003	Peter	2026-04-01	78	At risk customer
1004	Sarah	2026-02-15	123	Inactive customer
1005	Thabo	2025-12-20	180	Inactive customer